

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> The power (P) dissipated in a circuit can be calculated using the formula $(P = IV = I^2R = \frac{V^2}{R})$, where (P) is power, (I) is current, (V) is voltage, and (R) is resistance. Here we know the power $(P = 40)$ watts and the voltage $(V = 12)$ volts. We need to find the resistance (R) first. Since $(P = I^2R)$ and $(P = \frac{V^2}{R})$, we can equate the two expressions if the power can be expressed in terms of the resistance or voltage. Let's find (R) first: $(R = \frac{V^2}{P} = \frac{12^2}{40} = \frac{144}{40} = 3.6)$ ohms. Now we know the resistance, we can find the current using $(P = I^2R)$: $(40 = I^2 \times 3.6)$, solving this for (I) gives us $(I^2 = \frac{40}{3.6})$. Taking the square root of both sides, we find $(I = \sqrt{\frac{40}{3.6}} \approx 3.33)$ amps. </think>
<answer> E </answer>